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SEPTEMBER 2026

JANUARY 2019 FLIGHT DELAYS



WHERE DELAYS BEGIN, HOW THEY SPREAD,
AND IF THEY ARE HARD TO UNDO

DATA & DEFINITIONS

MAIN QUESTIONS

Where and when do delays begin?

Do they spread through an aircraft's day?

Can flights recover in the air?

Does traffic volume explain any of it?

583,985 January 2019 flight records,
346 airports, **17** carriers

DELAYED FLIGHT

A delayed flight departed at least 15 minutes late

RECOVERED FLIGHT

A recovered flight departed late but arrived on time

EXCLUSION THRESHOLD

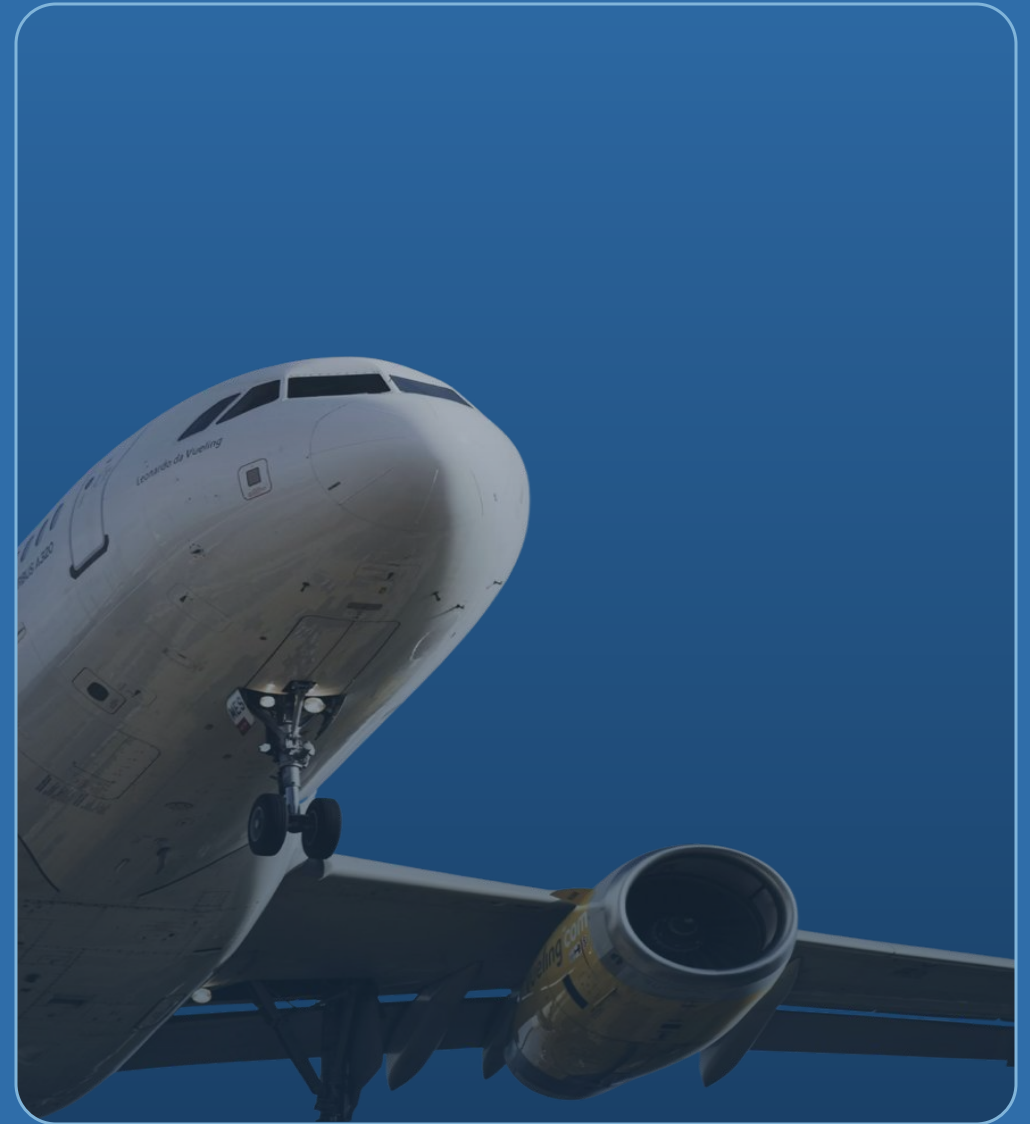
Airports under 1,000 January flights are excluded as unreliable

SECTION 01

01

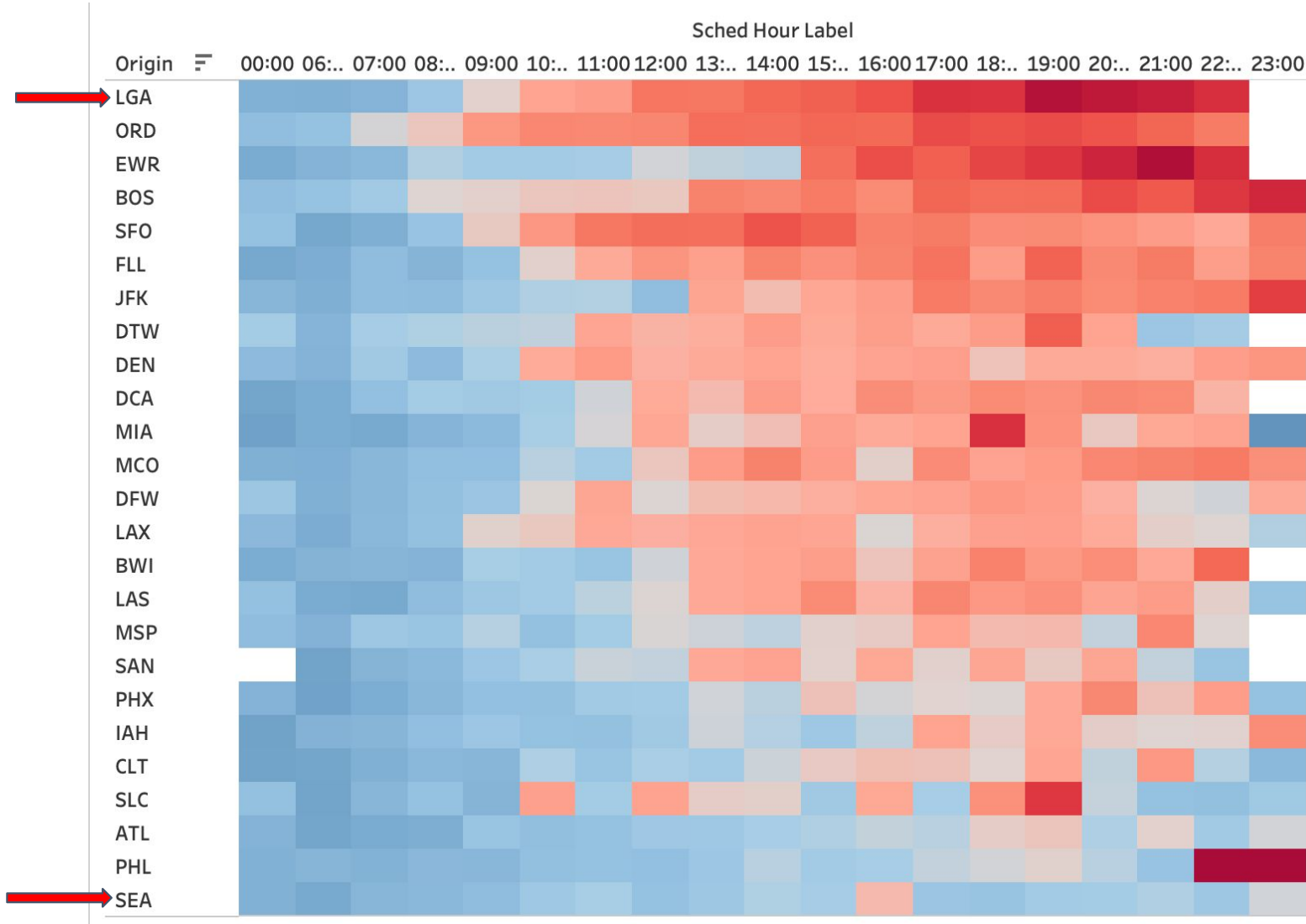
WHERE AND WHEN DELAYS BEGIN

PRESENTER: JACQUELINE PERALES



WHERE AND WHEN, IN ONE PICTURE

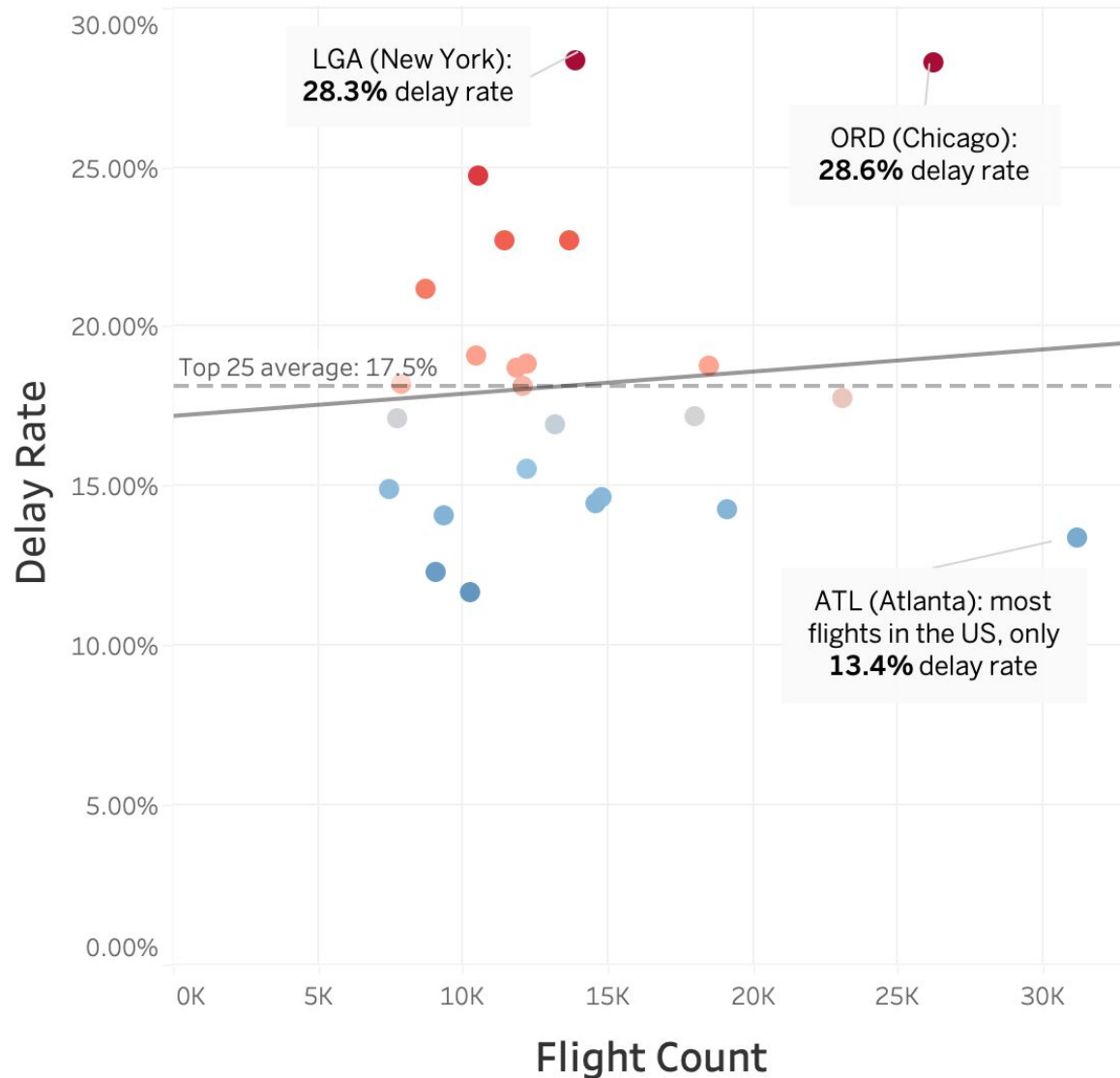
25 BUSIEST ORIGIN AIRPORTS BY SCHEDULED DEPARTURE HOUR · RED IS WORSE THAN THE 17.4% NATIONAL AVERAGE



WORST VS. BEST

- **LGA @ New York**
pale left, deepest red on the chart
- **SEA @ Seattle**
nearly the same colour all day

VOLUME DOES NOT CAUSE DELAY



ATLANTA

31,155 FLIGHTS

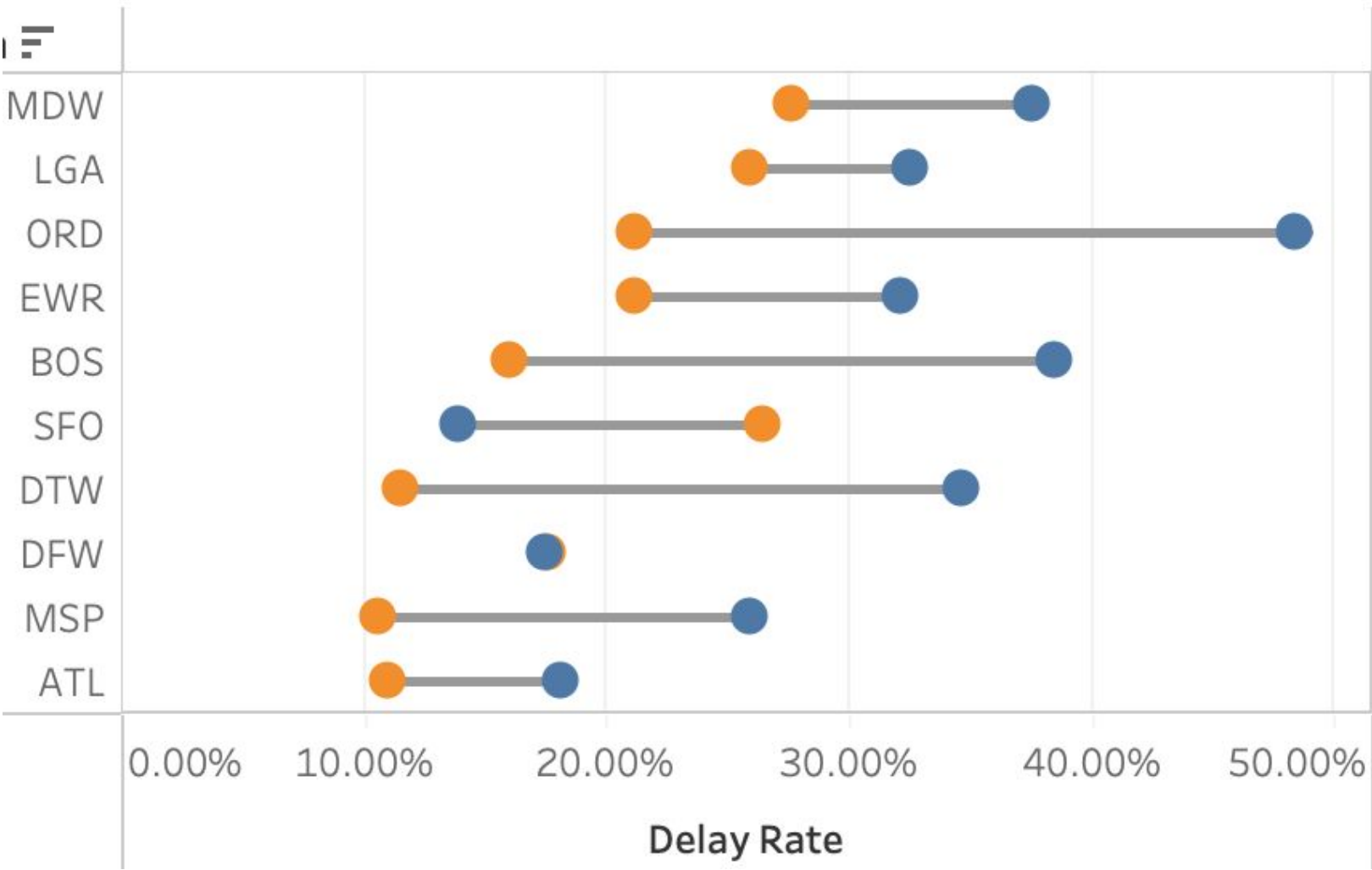
13.4% delayed, and never crosses 17.5% in any hour of the day

LAGUARDIA NEW YORK

13,872 FLIGHTS

28.31% delayed, less than half of atlanta's flights with almost 2x delay rate

WEATHER AFFECTED?



Weather Period

- Major storm days
- Normal days

LONGEST LINE vs SHORTEST LINE

	NORMAL	STORM CHANGE	
ORD delayed	21.4%	48.8%	+27.4
ORD cancelled	1.5%	30.4%	+28.9
DFW delayed	17.8%	17.6%	-0.2
DFW cancelled	1.2%	1.4%	+0.2

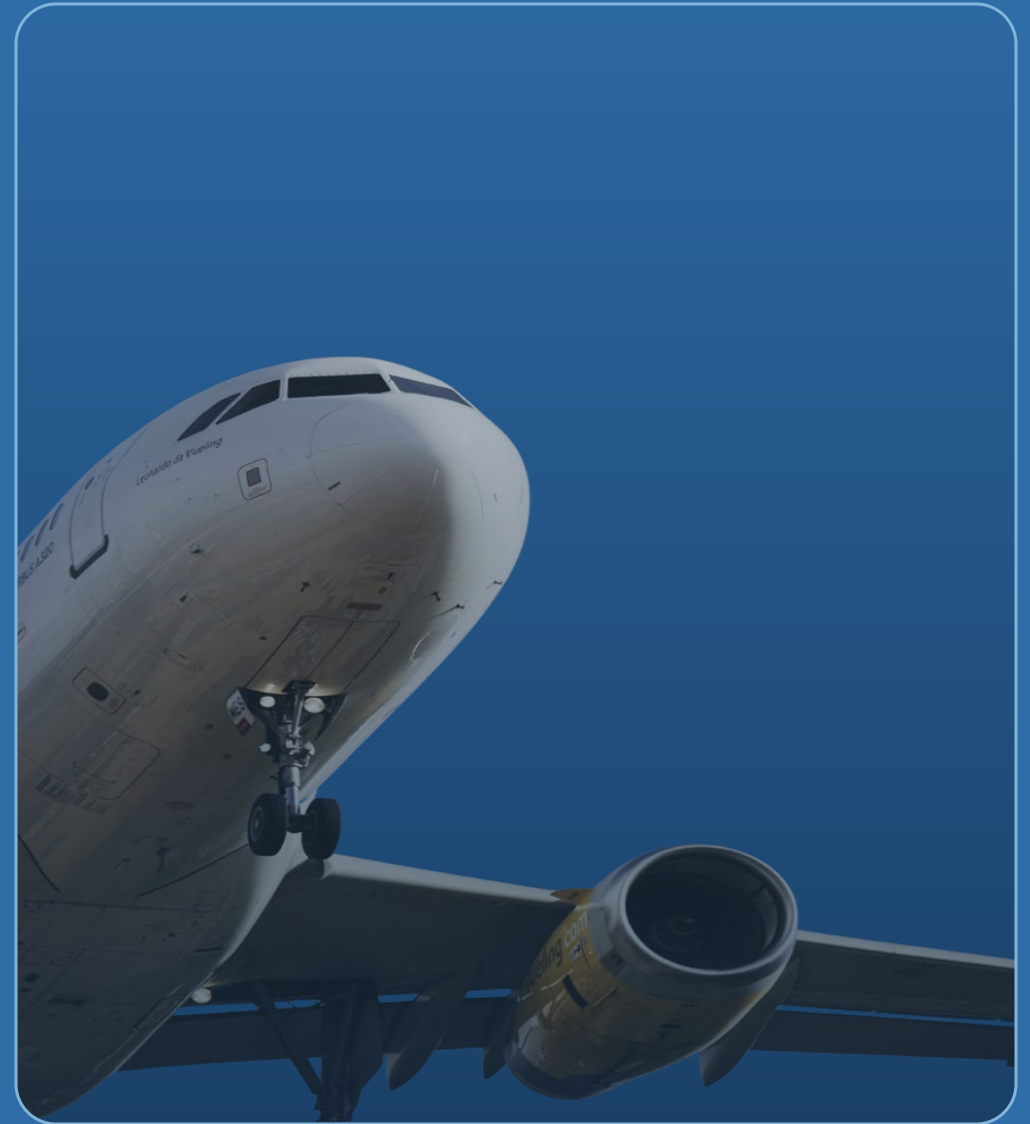
A LONG LINE MEANS JANUARY'S WEATHER/
STORMS HIT THAT AIRPORT. A SHORT LINE MEANS
THEY MISSED IT.

SECTION 02

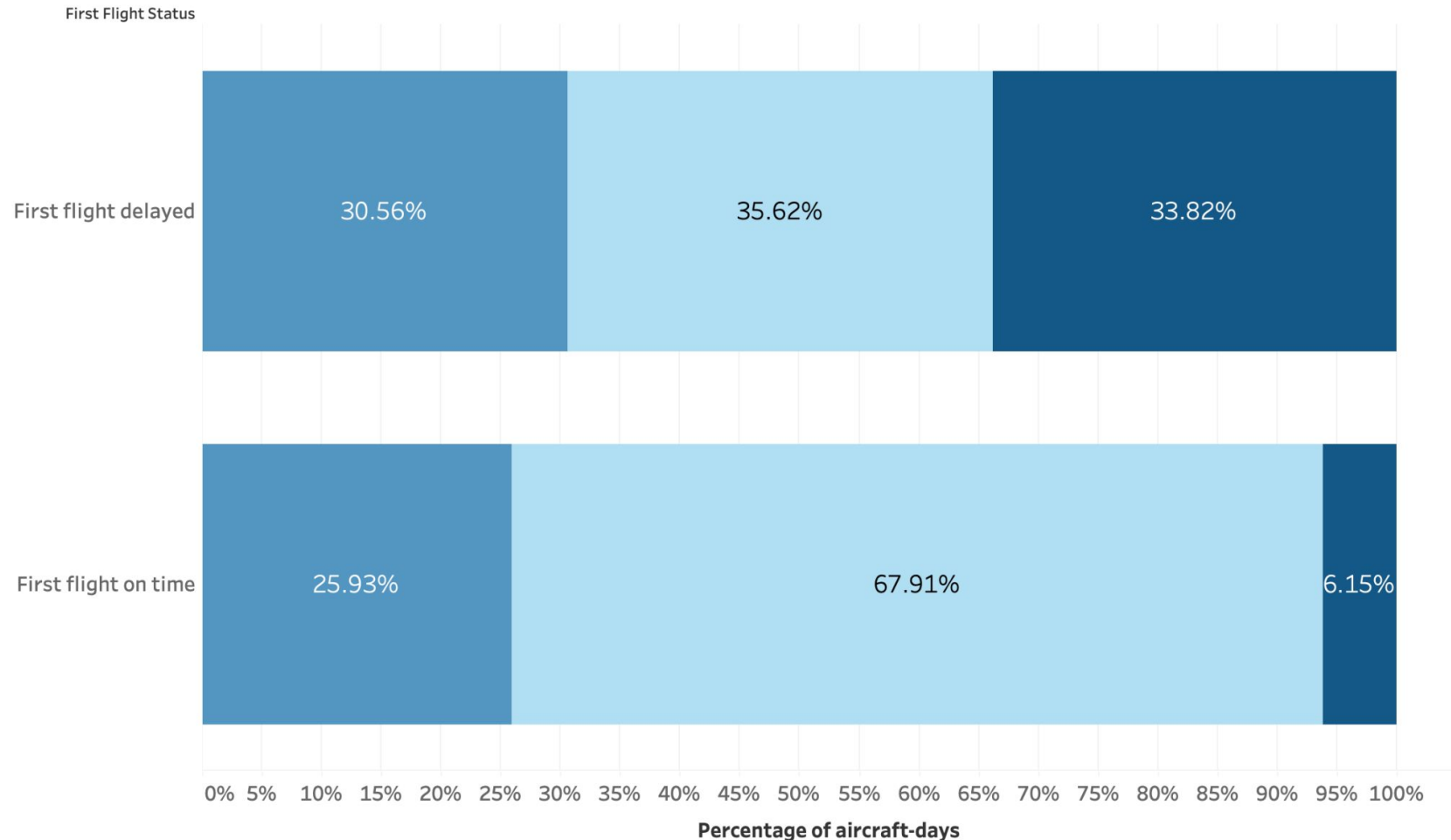
02

DOES DELAY TRAVEL WITH THE AIRCRAFT?

PRESENTER: Yael Abbo



WHAT HAPPENED TO THE AIRCRAFT'S LATER FLIGHTS?



64.38%

of aircraft-days that started late had at least one later delay

Only 33.82% had every later flight delayed

A late start usually damages the day. It does not usually destroy it.

■ All delayed
■ Some delayed
■ None delayed

DOES THE RISK CONTINUE?

Later flights stay at higher risk all day when the aircraft starts late.

46.5%

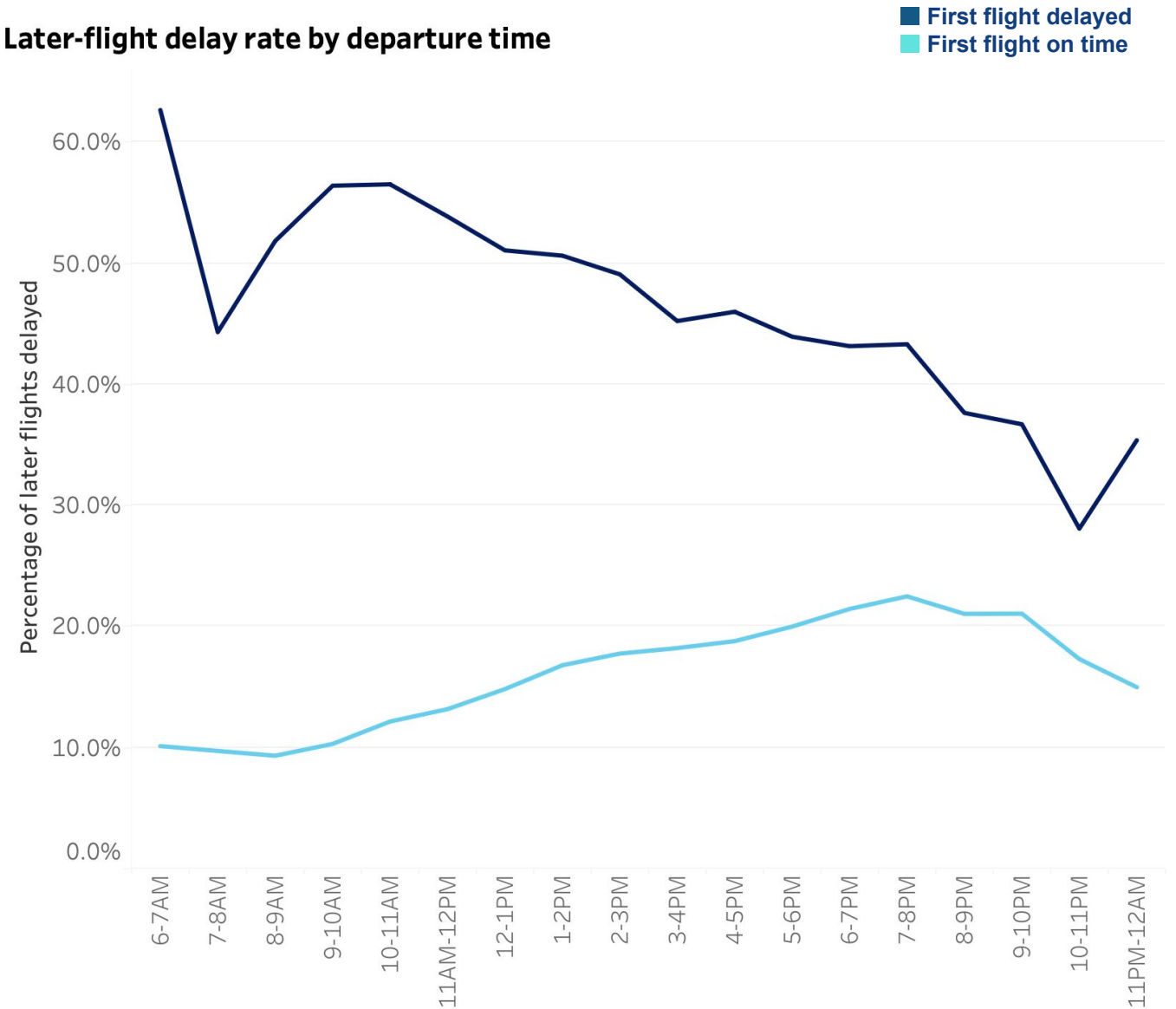
later-flight delay rate after a late first flight

17.1%

later-flight delay rate after an on-time first flight

The gap narrows through the day but never closes.

Later-flight delay rate by departure time

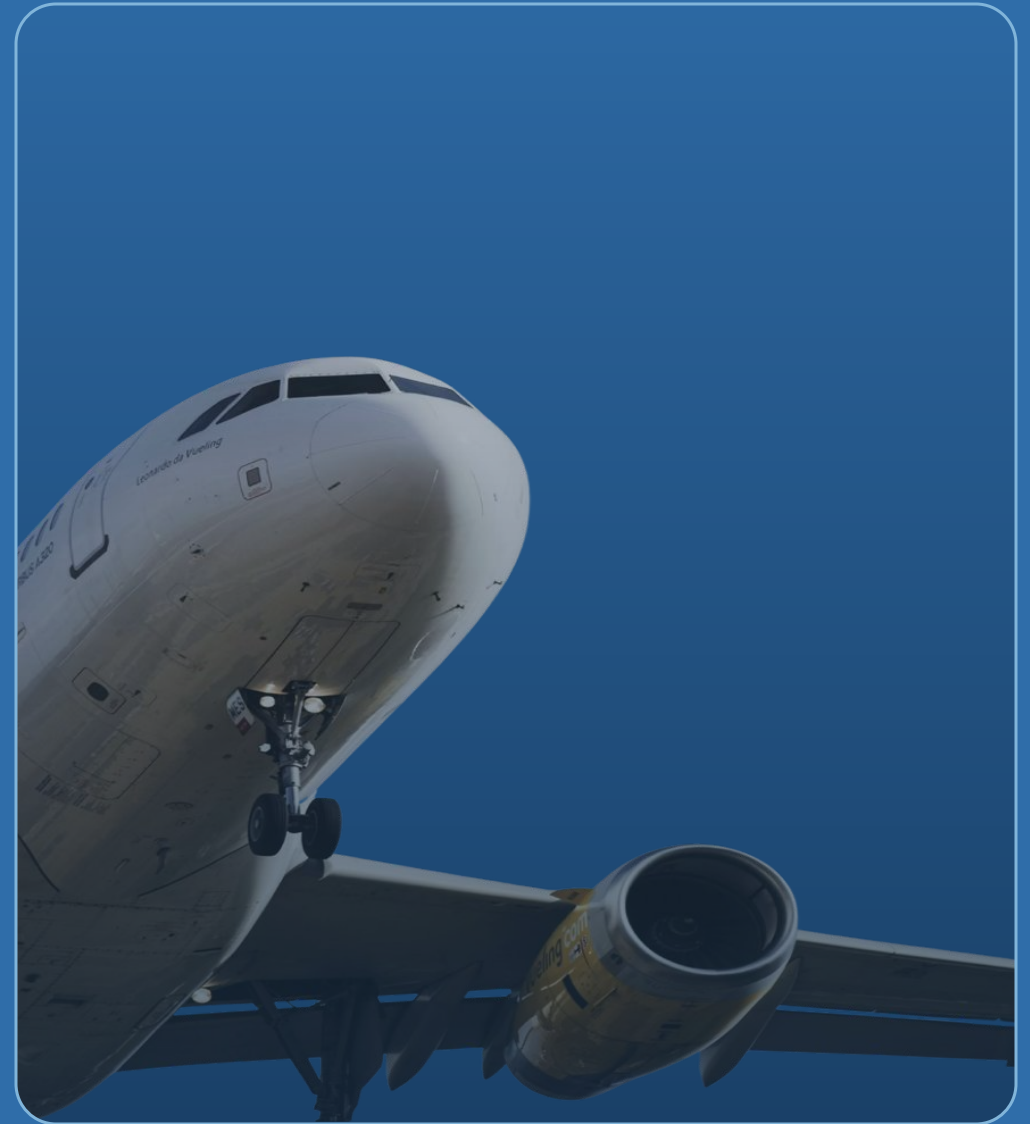


SECTION 03

03

CAN A LATE FLIGHT MAKE UP THE TIME?

PRESENTER: JOAO LUCAS



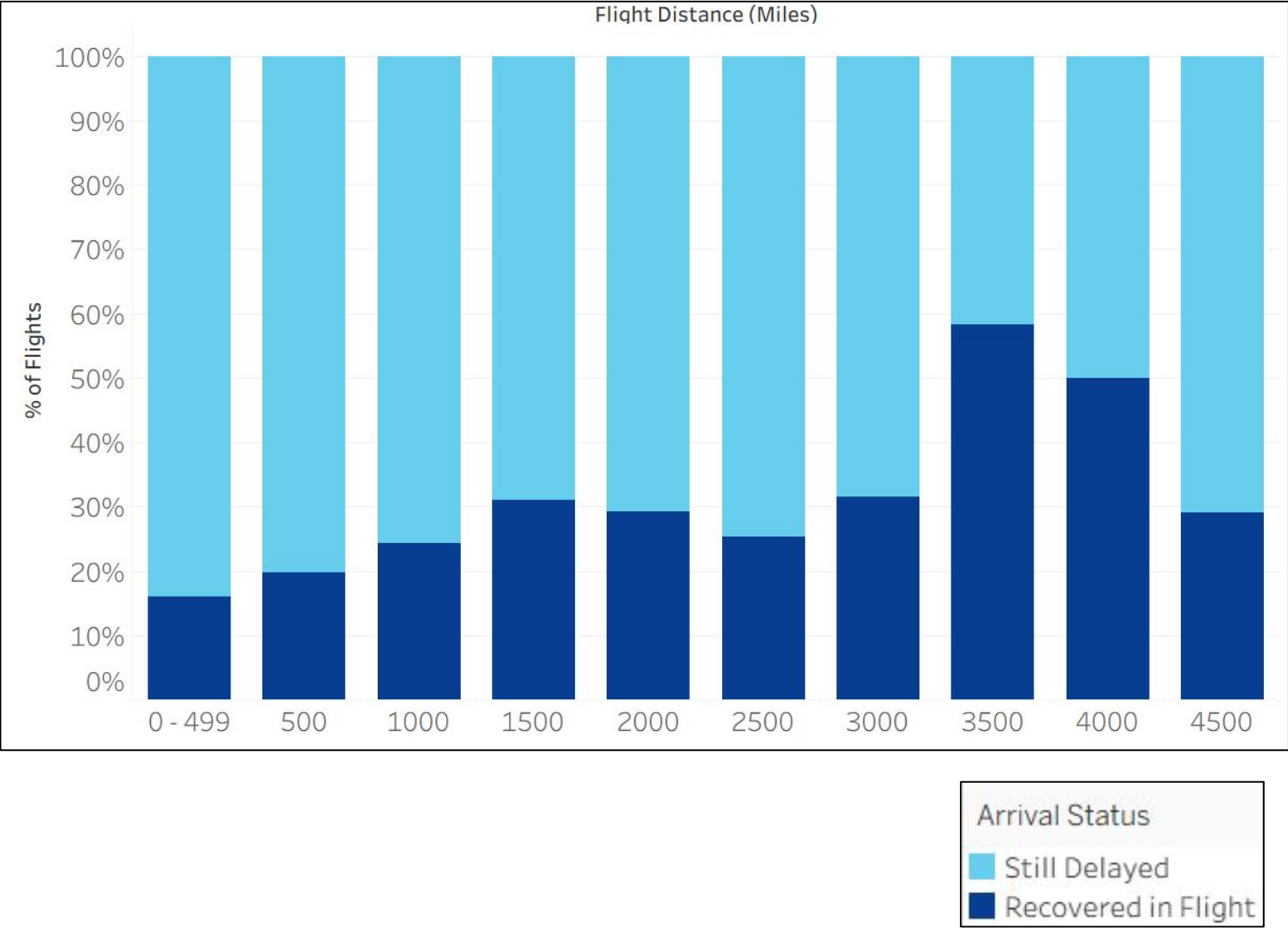
CAN FLIGHTS RECOVER? DISTANCE MATTERS.

THE LIKELIHOOD OF MAKING UP LOST TIME
RISES STEADILY WITH FLIGHT DISTANCE, BUT
NEVER BECOMES LIKELY.

20.4% of the 98,305 late departures still
arrived on time

UNDER 500 MI	15.9%
500 – 999 MI	19.7%
1,000 – 1,499 MI	24.3%
1,500+ MI	29.8%

Average growth: 4.63
percentage points

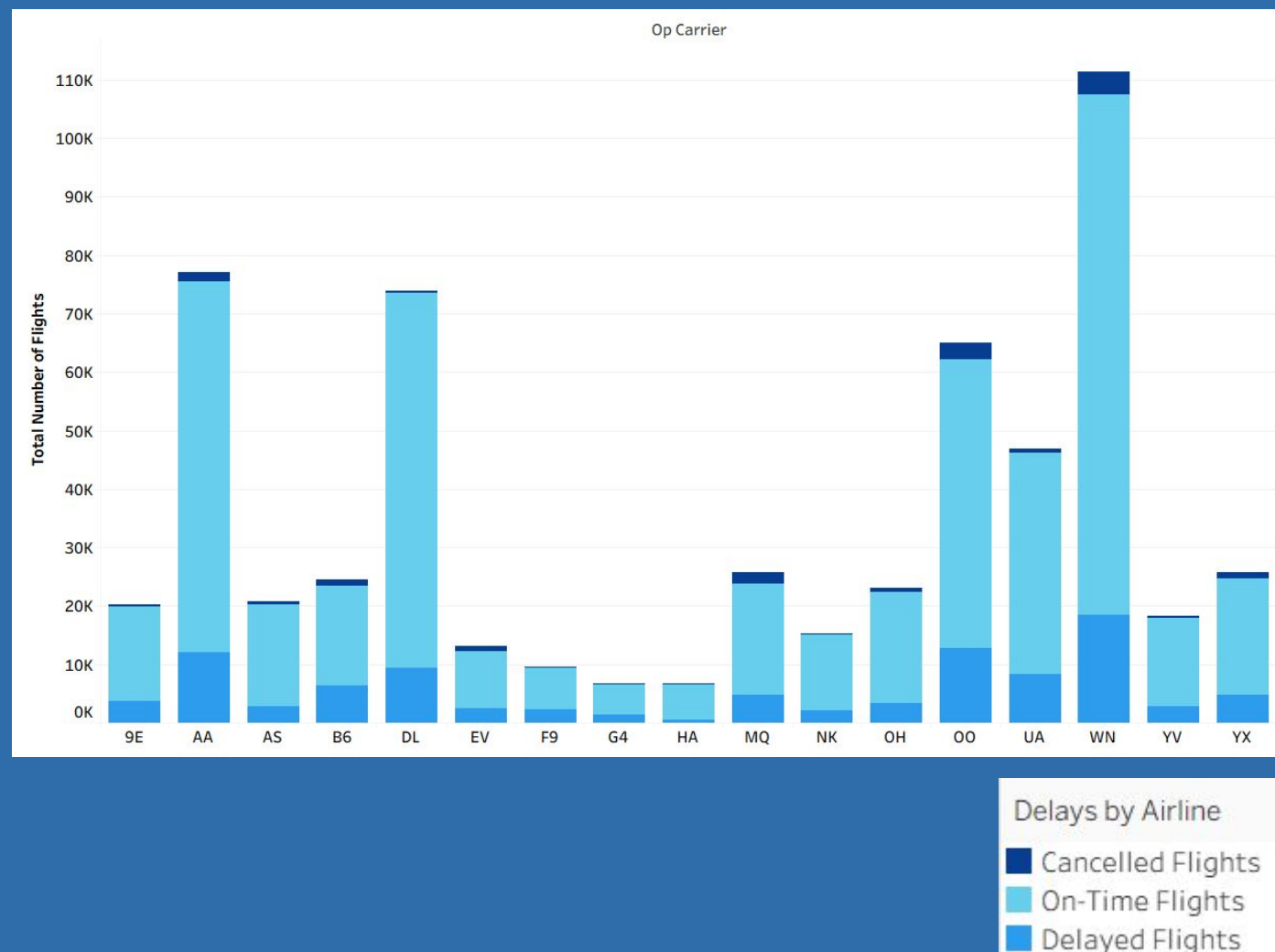


AIRLINE PERFORMANCE: WHO CANCELS THE MOST?

2.84% network-wide cancellation
rate across all 17 carriers

7.5% cancellation rate at MQ, the
clear outlier carrier

CANCELLATIONS AND DELAYS ARE PARTLY A CARRIER STORY, NOT ONLY A PLACE STORY. AN AIRPORT THAT CANCELS HEAVILY CAN LOOK PUNCTUAL ON DELAY RATE ALONE.



THE THREE ANSWERS DESCRIBE ONE SYSTEM

DELAY COMPOUNDS THROUGH THE DAY

7.0% at 6am, 24.6% by 7pm, and different airports fail in different ways.

BECAUSE IT TRAVELS WITH THE AIRCRAFT

One late first flight takes the rest of that plane's day from 17.1% to 46.5%.

AND FLYING CANNOT UNDO IT

Only 20.4% of late departures recover in the air, and just 15.9% under 500 miles.

* THE FIX HAS TO HAPPEN ON THE GROUND, IN THE MORNING.

WHAT WE WOULD TELL THE AIRLINE

01

PROTECT THE MORNING FLIGHTS ABOVE EVERYTHING

A 7am slot carries roughly 7% delay risk against 25% at 7pm, and a late first flight nearly triples the rest of that aircraft's day.

02

KNOW THE AIRPORTS ROOT ISSUE BEFORE SPENDING

MDW and LGA need turn-time and schedule changes on ordinary days. ORD, DTW, BOS and MSP need winter contingency.

03

COPY SUCCESSFUL AIRPORTS

The busiest airport in the country runs at 13.4% with a flat daily curve. Whatever ATL does about turn buffers already works.